# Session 1: Overview of computer and IT

FACULTY OF INFORMATION TECHNOLOGY
UNIVERSITY OF SCIENCE - HO CHI MINH CITY

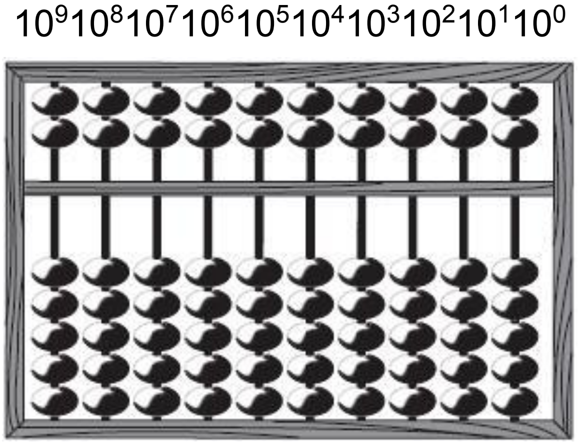
# Content

- History of Computing
- Categorization of Computers
- Hardware
- Software
- IT and Applications
- IT in Vietnam
- FIT-HCMUS

# HISTORY OF COMPUTING

# Computers

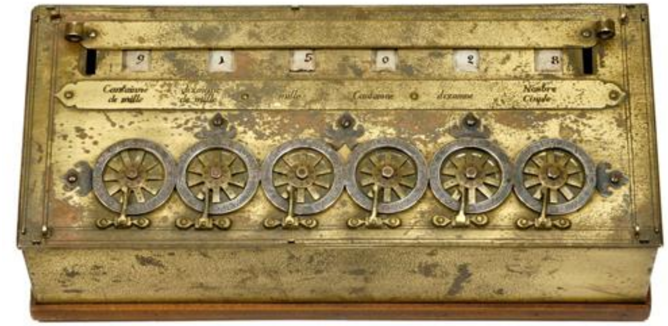
- The oldest computing device was the Roman abacus
- The most familiar version is the Chinese abacus.




Chinese abacus

# Computers

- In 1641, Blaise Pascal (1623 - 1662) built the first mechanical calculator for addition
- In 1671, Gottfried Leibritz (1646 - 1716) improved Pascal's calculator for addition, subtraction, multiplication and division.



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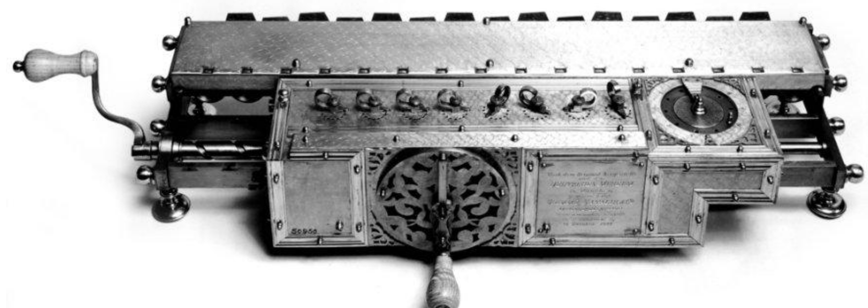


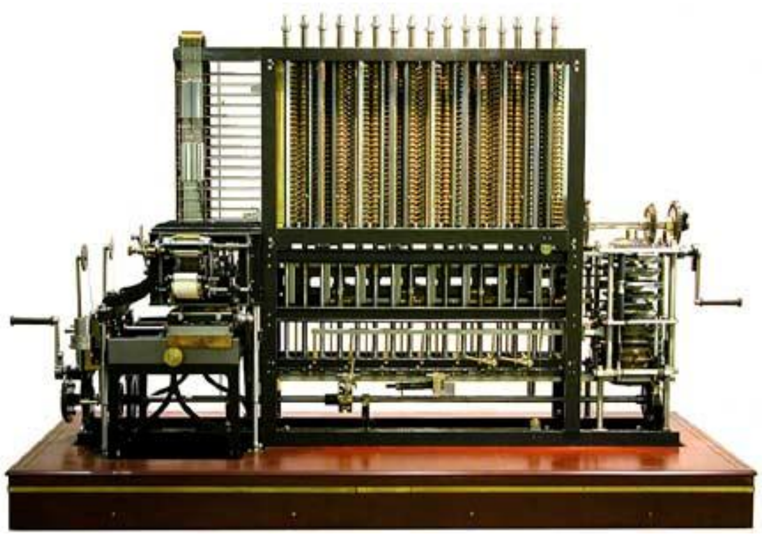
The mechanical calculator Of Gottfried Wilhelm Leibniz for addition, subtraction, multiplication and division.

*Nguồn : computerhistory.org*

# Computers

- In 1833, Charles Babbage (1792 - 1871) suggested that mechanical calculators should not be developed and proposed computers with external programs (on punched cards).

Charles Babbage



Charles Babbage 's computer

*Nguồn : wikipedia*

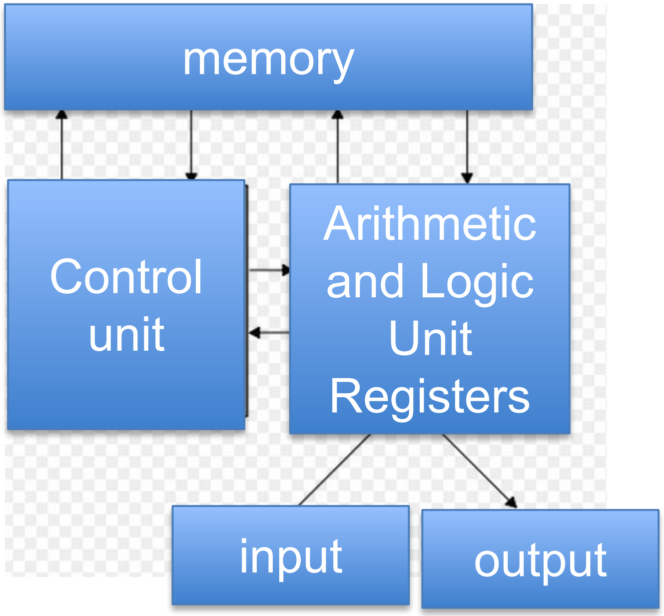
# Computers

- punched card


# Computers

- In 1945, John Von Neumann introduced a significant principle: the program must be stored in the computers and execute the program's instructions sequentially (one instruction at a time).

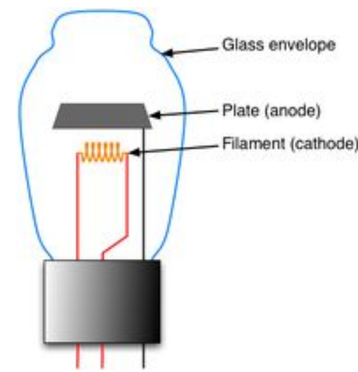
John Von Neumann

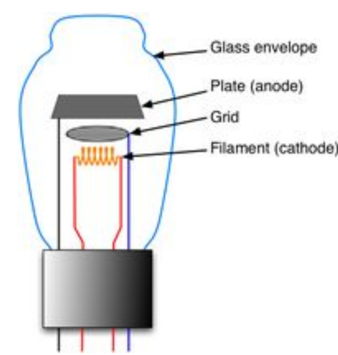



The Architecture of J.V. Neumann (1947)
https://vi.wikipedia.org/wiki/John_von_Neumann

# Generations

- First generation (1940s – 1950s)
  - Using vacuum tubes
  - ENIAC machine (USA) is 30.5m long, weighs 30 tons, 18000 vacuum tubes, uses punch cards, performs 1900 additions / second, serves for defense purposes (ballistic, making atomic bombs , ...)
  - UNIVAC machine is 10 times faster than ENIAC machine, using more than 5000 vacuum tubes



1946
ENIAC

Nguồn : computerhistory.org

1951
UNIVAC

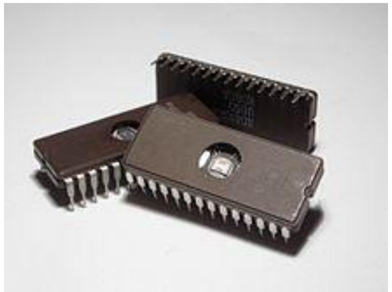
Nguồn : computerhistory.org

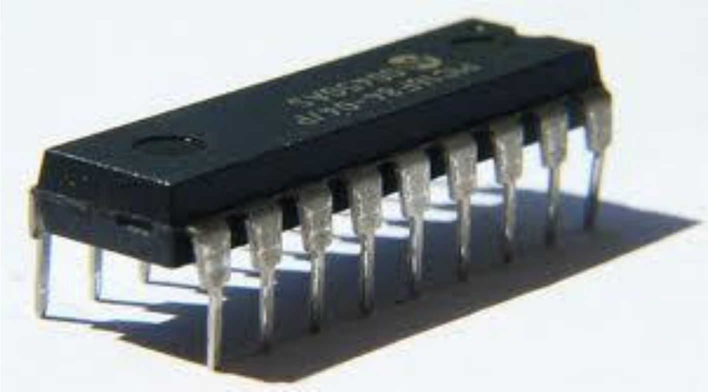
# Generations

- Second generation (1950s – 1960s)
  - Using semiconductors - transistors (smaller and cheaper, consume less power and generate less heat than vacuum tubes)
  - IBM 7090 reached 2 million calculations / second; participated in Mercury Project (USA) (put the first American astronaut in space); calculated the largest prime number at that time (1961) with 1332 digits
  - M-3, Minsk-1, Minsk-2 (Soviet Union)
  - NNLT: COBOL, FORTRAN

# Generations

- Third generation (1960s – 1970s)
  - Using Integrated Circuit (smaller, faster, cheaper, ...)
  - IBM360 (USA) performs 500,000 additions / second (250 times more than ENIAC)

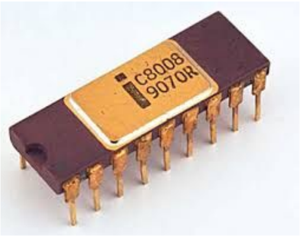




# Generations

- Fourth generation (1970s – 1990s)
  - Using large-scale integrated circuit (LSI) and very large-scale integrated circuit (VLSI)
  - Intel 4004 in 1971 (4-bit processor)
  - Intel 8008 in 1972 (8-bit processor)
  - Intel 8086 in 1978 (16-bit processor)
  - Intel Core i7 (1,170,000,000 transistors, 6 cores, 12 threads working simultaneously)
  - Other (Snapdragon 855, Apple A11 Bionic)
  - Parallel processing

Intel 4004



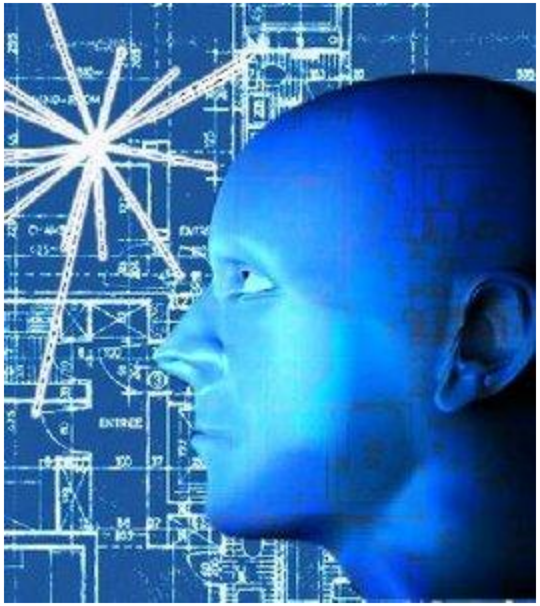



1973
Personal Computer - Micral N.
François Gernelle

1981
IBM PC

# Generations

- Fifth generation (1990s - now)
  - Working on artificial intelligence
  - Communicating directly with people in natural language,
  - Learning the knowledge
  - Expressing emotions ...



# Generations

- Sixth generation (near future?)
  - **Quantum Computer**
  - Uses **Qubits** instead of classical bits
  - **Advantages**:
    - Simulate molecules and materials
    - Optimize complex systems
    - Break or create cryptography
  - **Challenges**:
    - Requires near absolute zero environment
    - Still in experimental stage


INTERNET of
THINGS

# Virtual Reality (VR)

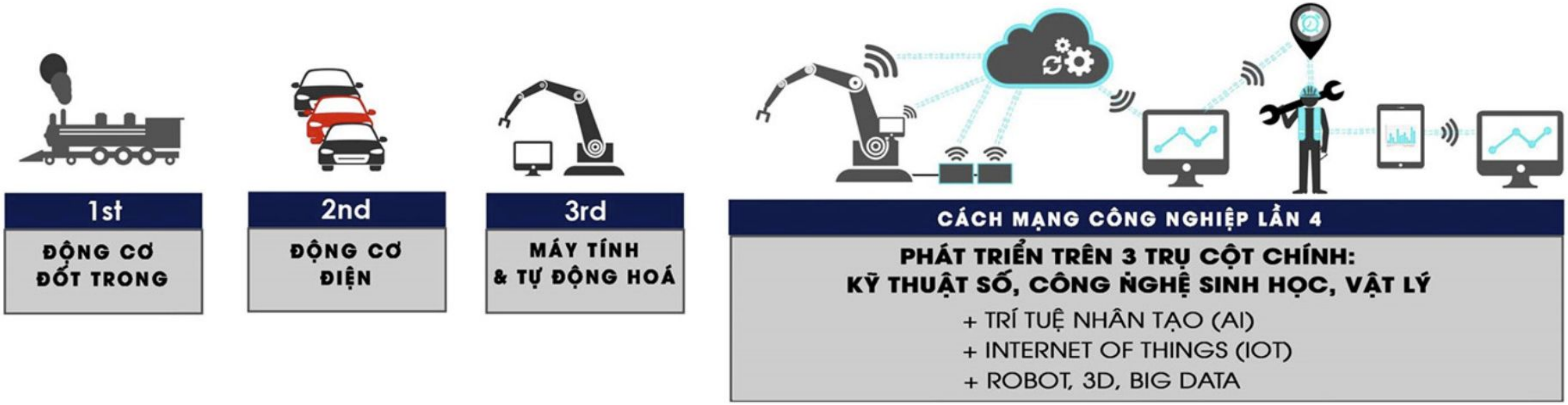
# Augmented Reality (AR)

# Hologram

# Industry 4.0

- The 1st industrial revolution marked by transition from hand production methods to machines through the use of steam power and water power.
- The 2nd revolution took place thanks to the application of electricity for mass production.
- The 3rd revolution used electronic and information technology to automate the production.
- Now, the 4th Industrial Revolution is a range of new technologies that are fusing the physical, digital and biological worlds, impacting all economies and industries, and even challenging ideas about what it means to be human. (Klaus Schwab)

# Industry 4.0





(Nguồn Internet)

# Components of Industry 4.0

- AI - Artificial Intelligence
- IoT - Internet of Things (Everything connected via the Internet)
- 3D technology: Virtual reality, Augmented reality, 3D printing
- Social networks, mobile networks, big data analytics, cloud computing (SMAC: Social, Mobile, Analytic, Cloud)

# Industry 4.0

- Which factors will play a decisive role in the 4.0 industrial revolution?

The answer is still ahead,
Maybe they could be inventions of IT!

# CATEGORIZATION OF COMPUTERS

# Supercomputer

- The most powerful today, it is integrated from hundreds to thousands of processors.
- Designed for simulations of global climate change, nuclear explosion, ...

# Mainframe

- Designed for multitasking
- Powerful input and output, focusing on processes of big data, such as financial transactions, insurance business, …

# Minicomputer

- Computers are between mainframes and microcomputers

# Microcomputer

- Computers are suitable for most users, including three main types:
  - Desktop
  - Laptop
  - Handheld

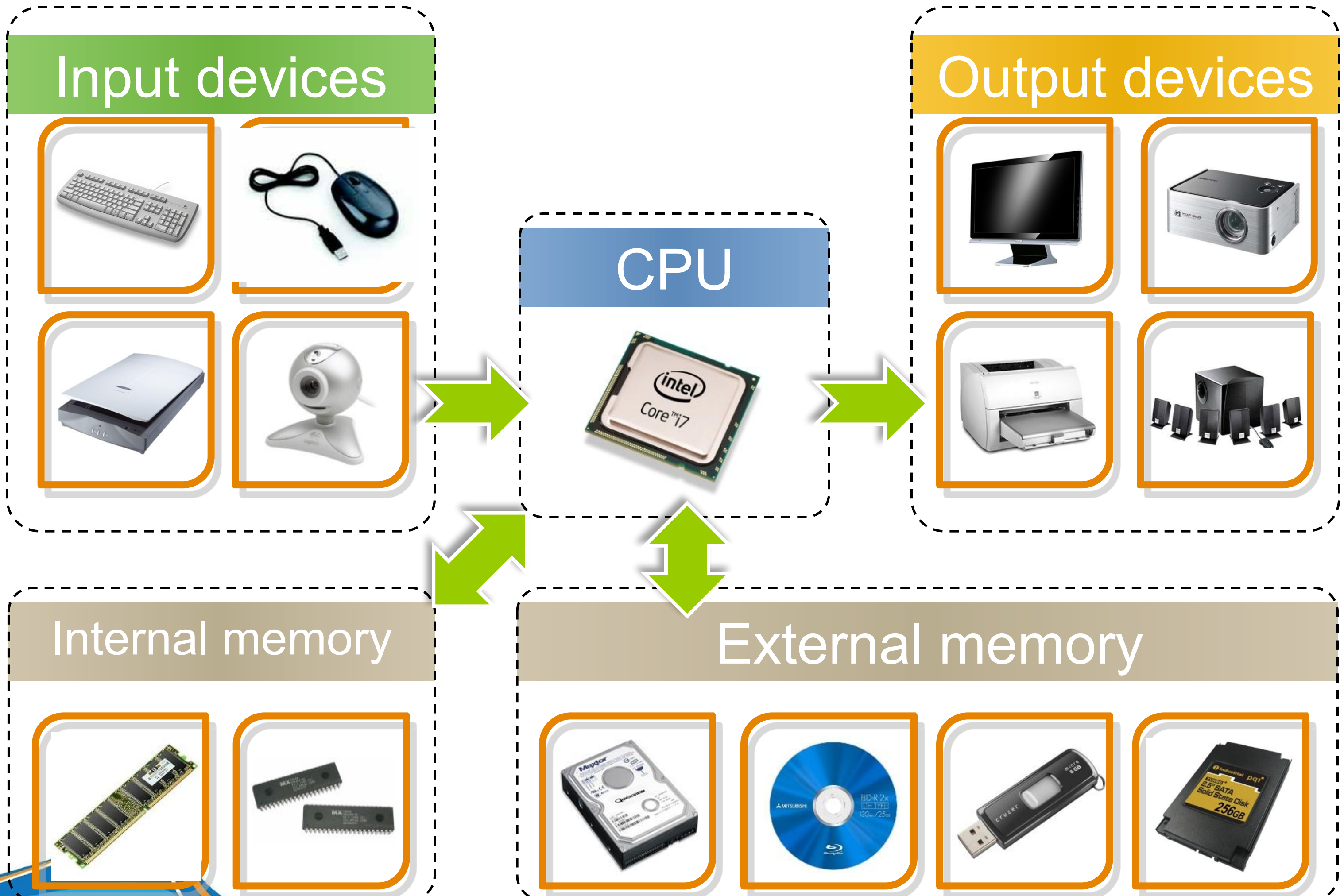
Desktop

Laptop

Handheld

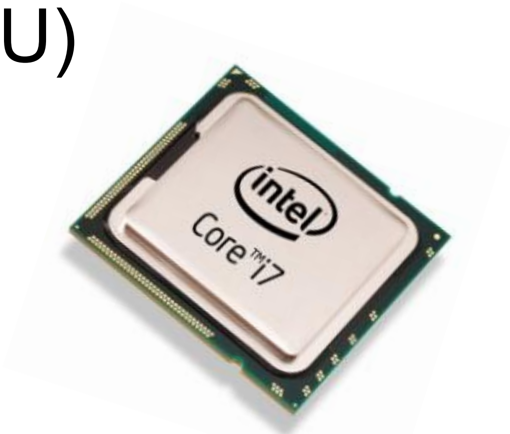
# COMPUTER ARCHITECTURE - HARDWARE

# Computer Architecture



# CPU

- Central Processing Unit - CPU
  - The CPU controls all computer operations.
- Include:
  - Control Unit (Control Unit - CU)
  - Arithmetic Logic Unit (ALU)
  - Registers
  - Bus line
  - Clock



Intel Core i7 CPU

# Internal memory

- ROM (Read Only Memory)
  - Storing the system program.
  - Data is always maintained even when the power supply is interrupted.


- RAM (Random Access Memory)
  - Storing data temporarily.
  - Data will be lost when power supply is interrupted

# External memory

- Advantages and disadvantages compared to internal memory
  - Advantages:
    - Storage capacity is larger
    - High reliability
    - Low cost
  - Disadvantages
    - Access speed is significantly slower

# External memory

- Categorization based on technical characteristics:
  - Magnetic system
  - Optical system
  - Flash drive
  - Solid state drive

# External memory

- Magnetic system
  - Tape: The first storage device, slow speed, often used to back up data.
  - Floppy Disk: slow speed, not long life.
  - Hard Disk: Multi-layer, up to TBs capacity, fast speed, long life.


# External memory

- Optical system
  - CD (Compact Disk): 700MB.
  - DVD disc (Digital Video / Versatile Disk): up to 17GB.
  - Some enhancements from DVD:
    - HD DVD / Blu-ray (30 / 50GB)
    - HVD (500GB up to 3.9TB)
    - 5D DVD (10TB)

# External memory

- Flash Drive
  - Developed over the last 10 years, eliminating the mechanical properties of magnetic and optical discs.
  - The small size, convenient communication via USB (Universal Serial Bus) so it has made the floppy disk no longer exist.
  - Common capacity ranges

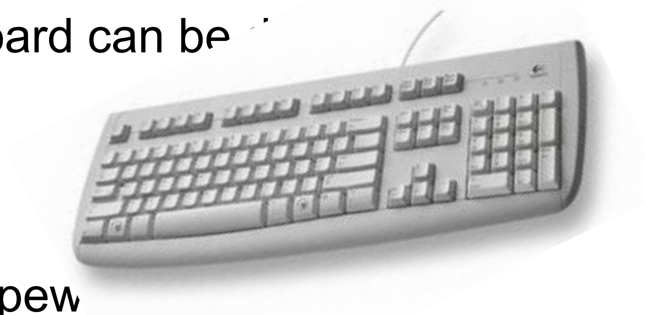
  from 8 GB to 32 GB.

# External memory

- Solid State Drive
  - Using solid memory to store data.
  - The read speed is 3 times faster, the write speed is 1.5 times faster than the normal hard drive.
  - Low power consumption, suitable for mobile devices.
  - The price is higher than a normal hard drive.
  - The largest capacity in 2010 is 1 TB and costs about $ 2,200.
  - 1 TB (10-2015) -> 300-400 $
  - 500 GB (08-2018) -> 100 $

# Input Devices

- Keyboard: Standard input device. The keys on a keyboard can be d
several groups based on their functions:



  - Typing (alphanumeric) keys.

   These keys include the same letter, number,

   punctuation, and symbol keys found on a traditional typew

  - Control keys. These keys are used alone or in combinati other keys to perform certain actions. The most frequently used control keys are Ctrl, Alt, the Windows logo key Picture of the Windows logo key, and Esc.
  - Function keys. The function keys are used to perform specific tasks. They are labeled as F1, F2, F3, and so on, up to F12. The functionality of these keys differs from program to program.
  - Navigation keys. These keys are used for moving around in documents or webpages and editing text. They include the arrow keys, Home, End, Page Up, Page Down, Delete, and Insert.
  - Numeric keypad. The numeric keypad is handy for entering numbers quickly. The keys are grouped together in a block like a conventional calculator or adding machine

# Input Devices

- Mouse: A hand-sized device to move the mouse cursor.
- Scanner: Convert documents into digital images.

Mouse

Scanner

# Input Devices

- Webcam & Camera: Record images from the real world to your computer.
- Digital Camera: Capture images from the real world and put them on the computer.

Webcam

Digital camera

# Input Devices

- Drawing Tablet: Use a stylus to draw on the electronic board
- Barcode Reader (Barcode Reader): Used to read barcodes (encrypted numbers).

Drawing Tablet

Barcode Reader

# Output Devices

- Monitor: Standard output device
  - Common types: CRT and LCD.
  - Resolutions: 800x600, 1024x768, ...
  - Sizes: 15 ", 17", 19 ", 22" ...


CRT

LCD

# Output Devices

- Projector
- Printer
- Speaker

Projector

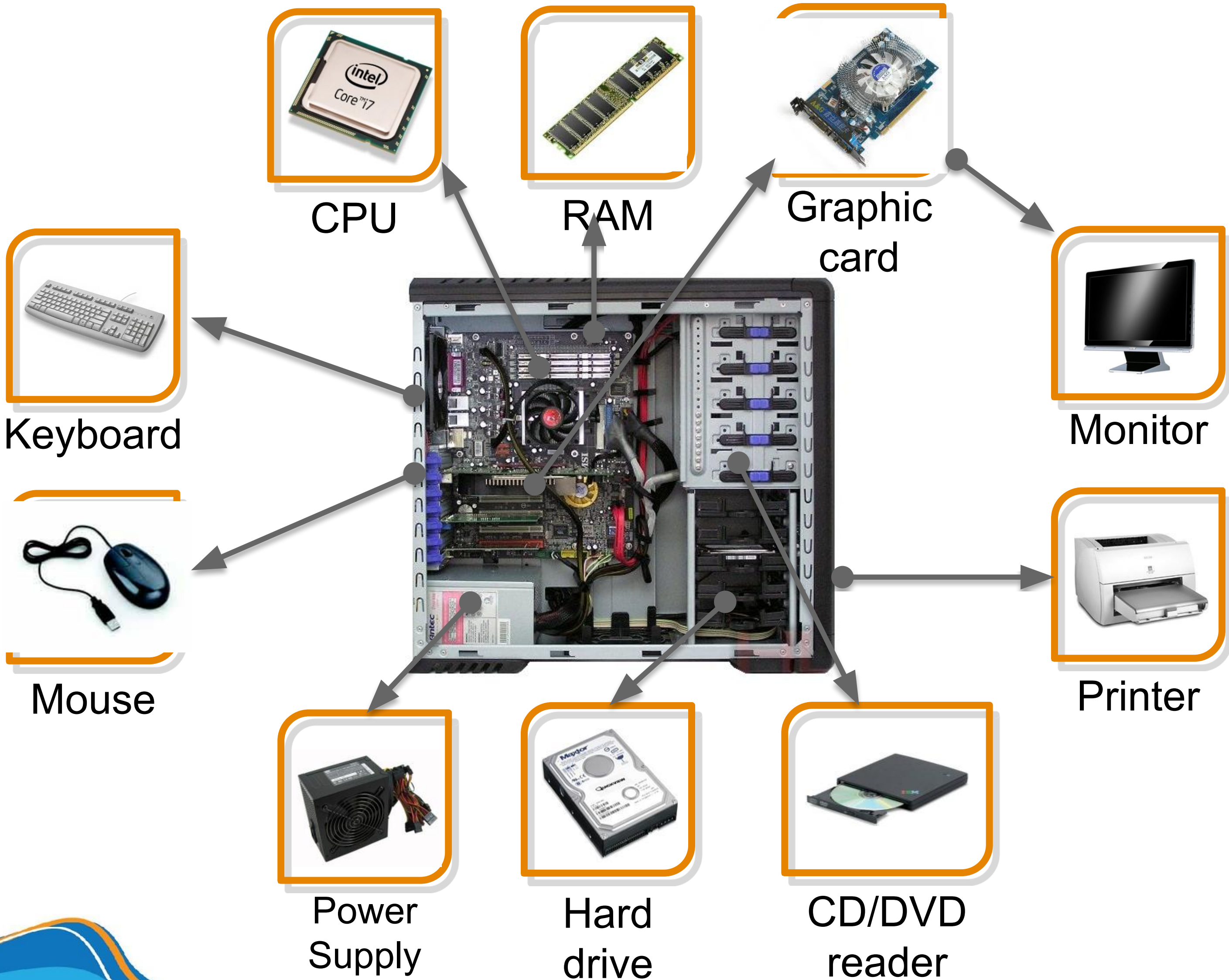
Printer

Speaker

# Motherboard

- Motherboard plays an important role and is a bridge for the devices.
- There are many devices mounted on the motherboard such as: computer power, CPU, RAM, control board (graphics, audio, network), hard drive, disk reader (CD, floppy disk), monitor, keyboards, mouse, ...

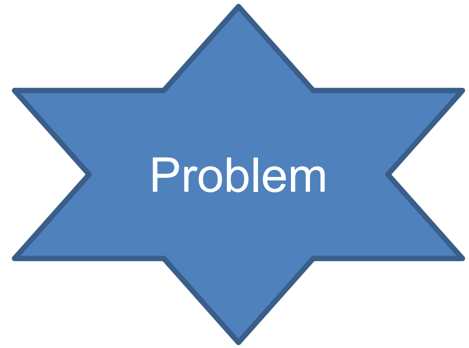
# Inside a PC case



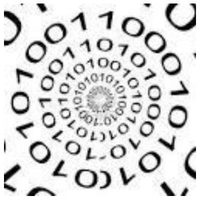
# SOFTWARE

# Definition

Software is a set of instructions written in a programming language with a syntax to automatically perform certain tasks or solve a certain problem.

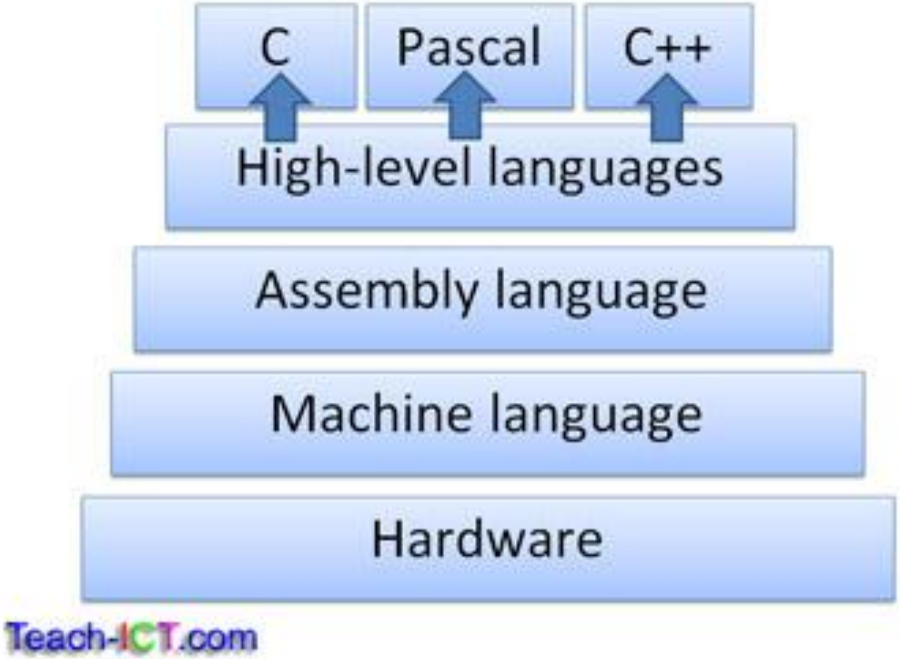



If .... Then ...else
Begin
... End
...



# Definition


Simula
Visual
C++
Visual Basic
APC
HTML
PL/I
Oberon
Algol
Cobol
Eiffel
Fortran
Basic
Prolog
awk
ADA
Icon
Lisp
Pascal
C++
sed
C
Modula-2
Smalltalk
Snobol
perl
Scheme
Delphi
JAVA


C
Pascal
C++
High-level languages
Assembly language
Machine language
Hardware
Teach-ICT.com

# Programming Language Rankings

## TIOBE Index and RedMonk

| Ranking | 2018 | 2017 |
|---|---|---|
| 1 | JavaScrip | JavaScrip |
| 2 | Java | Java |
| 3 | Python | Python |
| 4 | PHP | PHP, C++ |
| 5 | C# | |
| 6 | C++ | |
| 7 | CSS | CSS, Ruby |
| 8 | Ruby | |
| 9 | C | C |
| 10 | Swift | Objective-C |

# Software Classification

- System software
  - Operating system (OS): Compile, communicate with hardware, communicate with users, manage resources, control devices, etc.
  - Windows, Linux, MacOS

Microsoft Windows


Fedora

MacOS

# Software Classification

- System software
  - Network software.
  - Database management software (system).

# Software Classification

- System software
  - Software for controlling peripheral devices (drivers).
    - Drivers are integrated into the system
    - Allow programs to interact devices
    - Each device may have more drivers corresponding to different operating systems

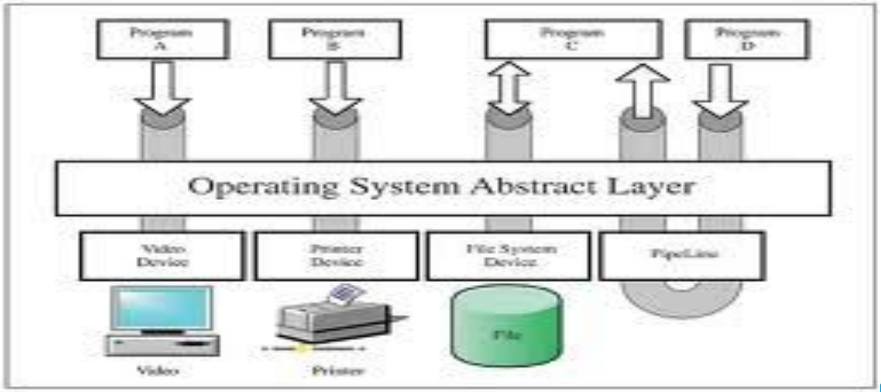


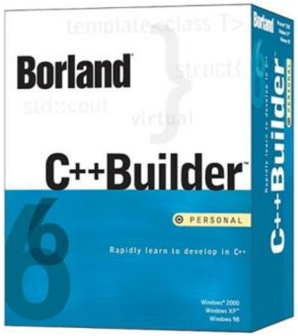

Figure 1. Device driver and file access. There are four programs (A, B, C, and D) reading data (C), writing data (A, B, D) or both (C). The arrows specify the data flow direction.

# Software Classification

- Software supports developing tools/software
  - Compiler, Interpreter.
  - Debugger.
  - Linkers, Loader.


Microsoft Visual Studio


Eclipse

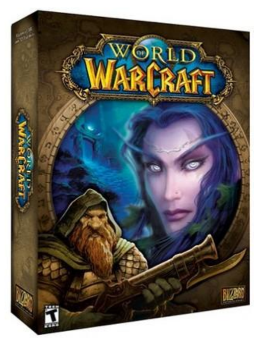



Borland C++ Builder 6

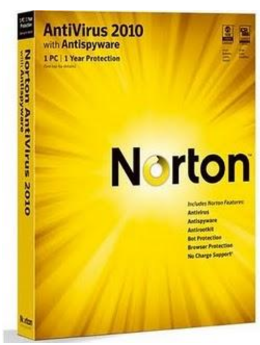
# Software Classification

- Application software
  - Working: office applications, business applications, graphic design, ...
  - Entertainment: games, listening to music, watching movies, ...
  - Utility : antivirus, data compression, ...


Microsoft Office



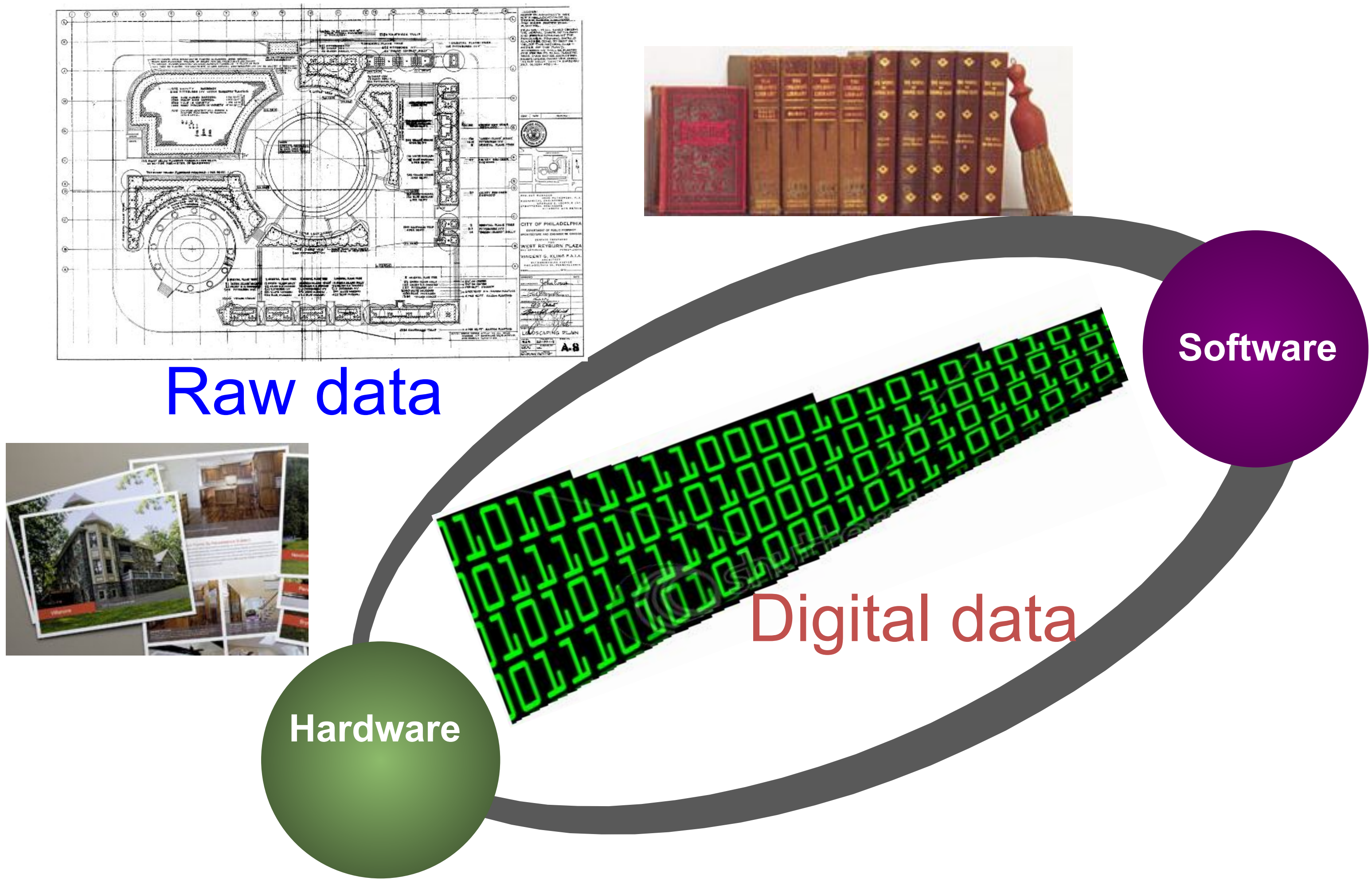

World of Warcraft




Norton Antivirus

# IT AND APPLICATIONS

# IT and applications



# Recording & digitizing data

- Data
- Text
- Image
- Video
- Audio

- Digital devices

**Recording**

- Data file
- Text file
- Image file
- Video file
- Audio file

**Digitizing**

manual way

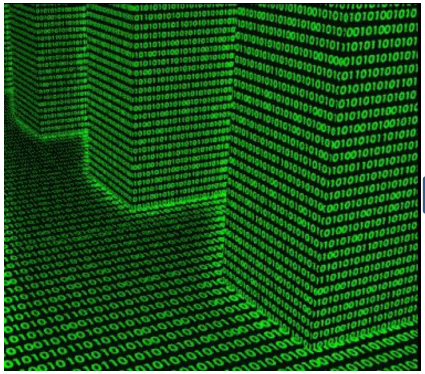
semi-automatic way

automatic way

# Demand for data

- Demand for data storage
- Demand for data search
- Demand for data extraction
- Demand for data visualization
- Demand for data transmission
- Demand for data sharing
- Demand for data security

# Data storage



# Data search

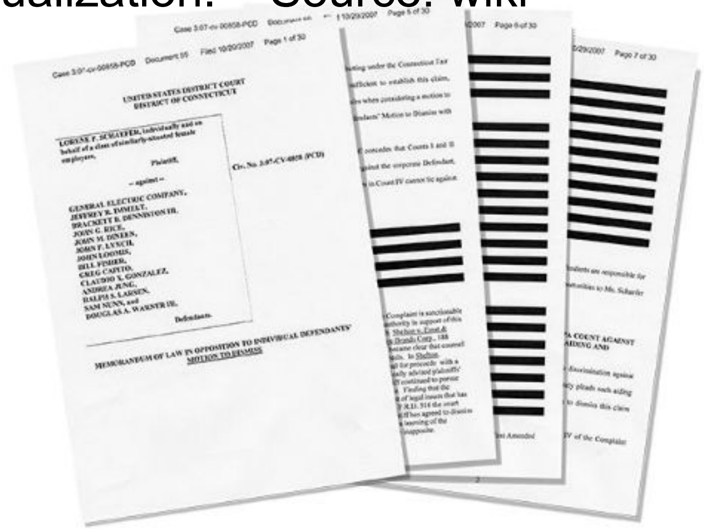
and

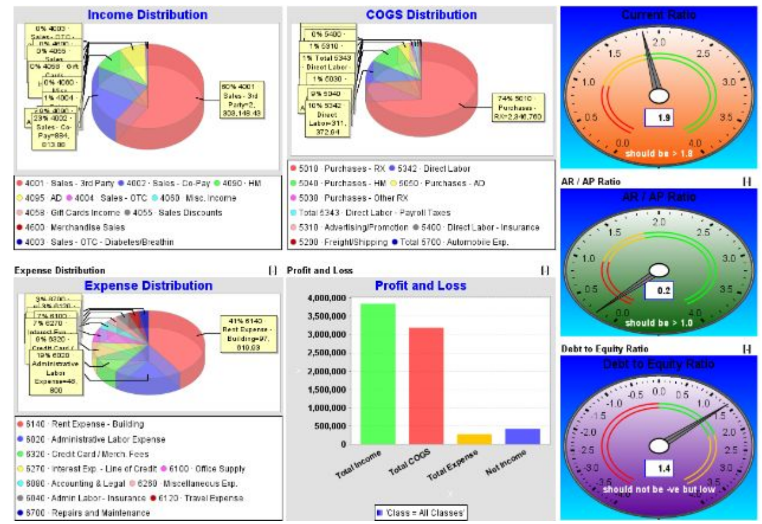
# Data extraction

**Data extraction** is the act or process of retrieving data out of (usually unstructured or poorly structured) data sources for further data processing or data storage (data migration) – source: wiki

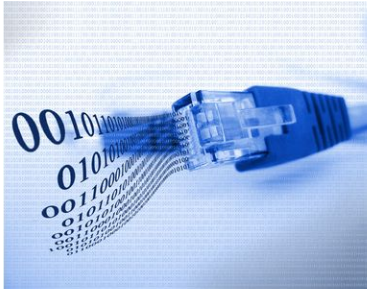
# Data visualization

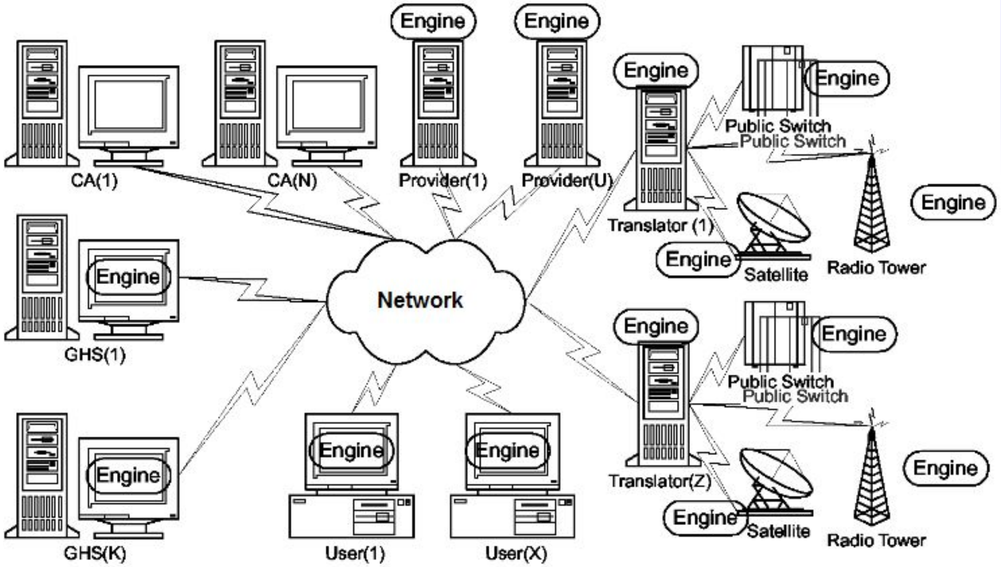
**Data visualization** is the graphic representation of data. It involves producing images that communicate relationships among the represented data to viewers of the images. This communication is achieved through the use of a systematic mapping between graphic marks and data values in the creation of the visualization. – Source: wiki





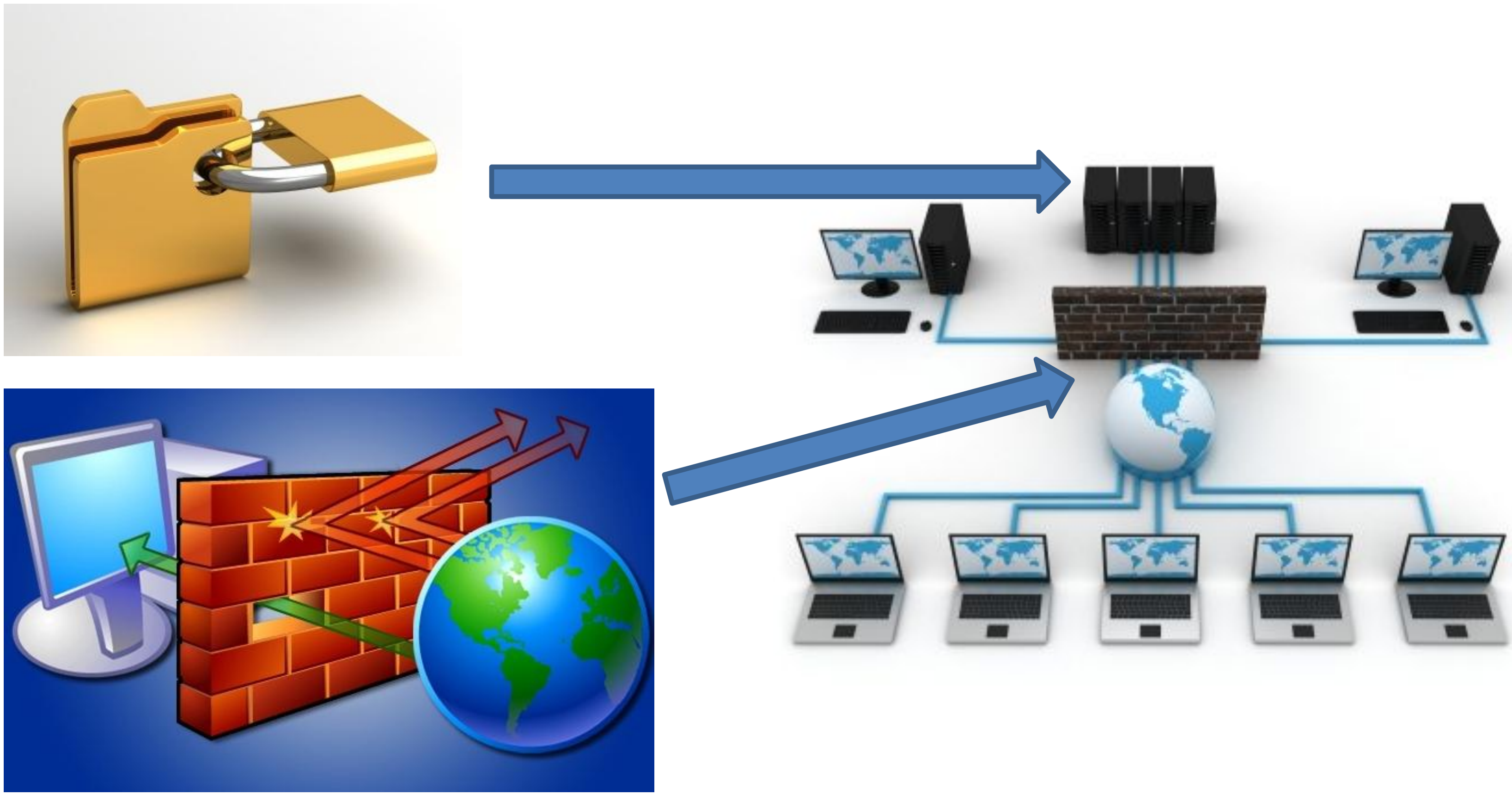

# Data transmission





# Data sharing

# Data security



# INFORMATION TECHNOLOGY (IT) IN VIETNAM

# IT in Vietnam

- Before 1975
  - 1964-1975: The South has the Computer Center used for the US military during the Vietnam War, using the US IBM 360 computer system.
  - 1968-1975: The North has Mathematics Department, using the Minsk-22 computer system of the Soviet Union (Russia).

# IT in Vietnam

The US IBM 360 computer system


The Minsk-22 computer system of the Soviet Union (Russia).

# IT in Vietnam

- After 1975
  - 1976: The Institute of Computational and Control Sciences was established in Hanoi, later renamed to Vietnam IT Institute.
  - 1988: Vietnam Association for Information Processing (VAIP) was established.
  - 1997: Vietnam officially connects to the internet worldwide.
  - 2002: Vietnam Software Association (VINASA) was established.

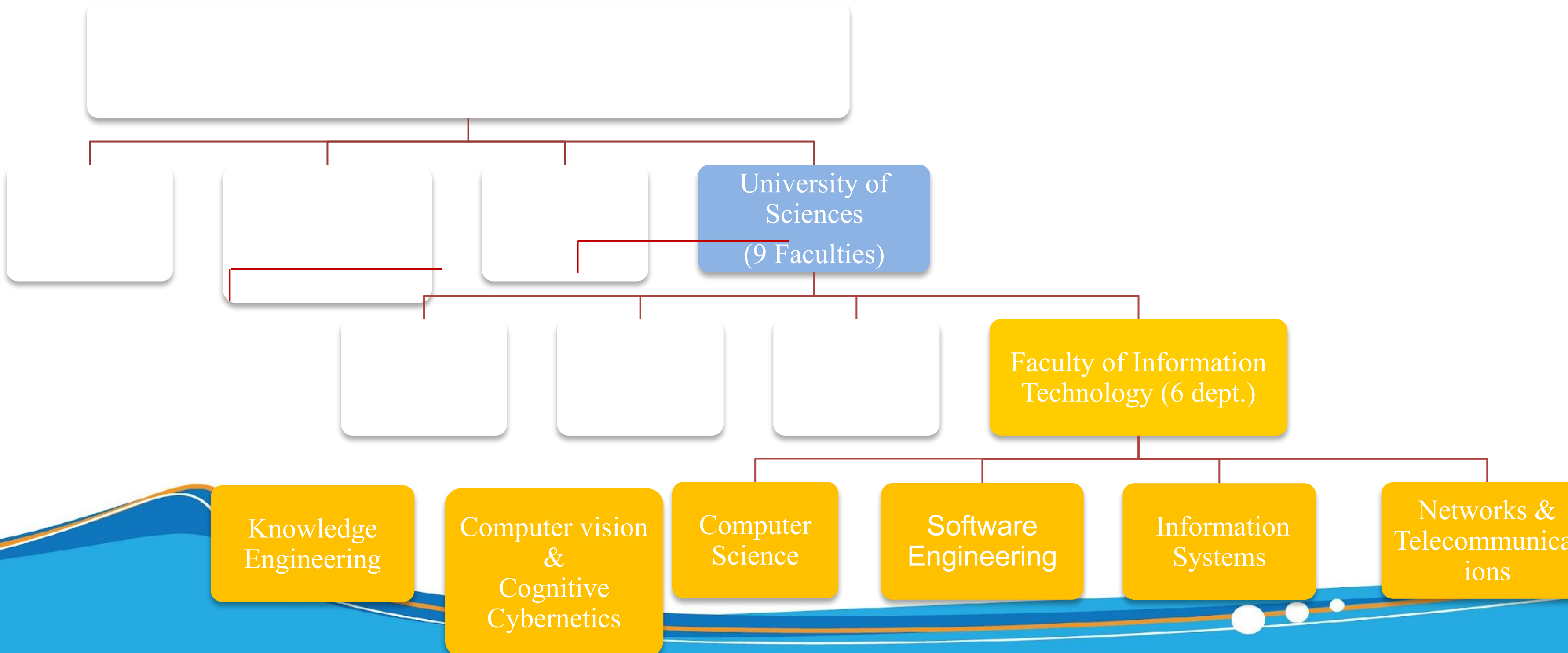
# IT in Vietnam

| Chỉ số | Số liệu | Nguồn / Ghi chú |
|---|---|---|
| **Mức lương trung bình ngành IT** | Khoảng **43,7 triệu VND/tháng** | Theo ITSS (năm 2024–2025) — mức lương trung bình ngành IT đạt 43,7 triệu, tăng 27,9% so với năm trước (Facebook) |
| **Mức lương theo cấp độ & kinh nghiệm** | • Người mới (freshers): 7 – 11 triệu VND/tháng • 3 năm kinh nghiệm: 20 – 32,5 triệu VND/tháng • Quản lý cấp cao hoặc chuyên gia: có thể vượt 30 – 50 triệu VND/tháng | Theo báo cáo JobOKO / các bài viết năm 2025 (vieclam.uet.vnu.edu.vn) |
| **Mức lương theo vị trí cao nhất** | CTO / CIO / VPoE: ~ 96 triệu VND/tháng | Theo bài đăng về báo cáo năm 2024–2025 của ITviec (Facebook) |
| **Nhu cầu nhân lực công nghệ** | Cần khoảng **500.000 nhân lực công nghệ** để đáp ứng xu hướng AI, cloud, an ninh mạng vào năm 2025 | Theo bài viết trên Nucamp về “Most in Demand Tech Job in Viet Nam 2025” (Nucamp) |
| **Thị trường việc làm & tăng trưởng ngành IT** | Nhiều dự báo cho thấy CNTT / công nghệ là ngành dẫn đầu nhu cầu việc làm tăng trưởng mạnh trong năm 2025 | Theo bài viết “Vietnam’s job market to surge in 2025, tech to lead the way” (Staffing Industry Analysts) |

# FIT-HCMUS

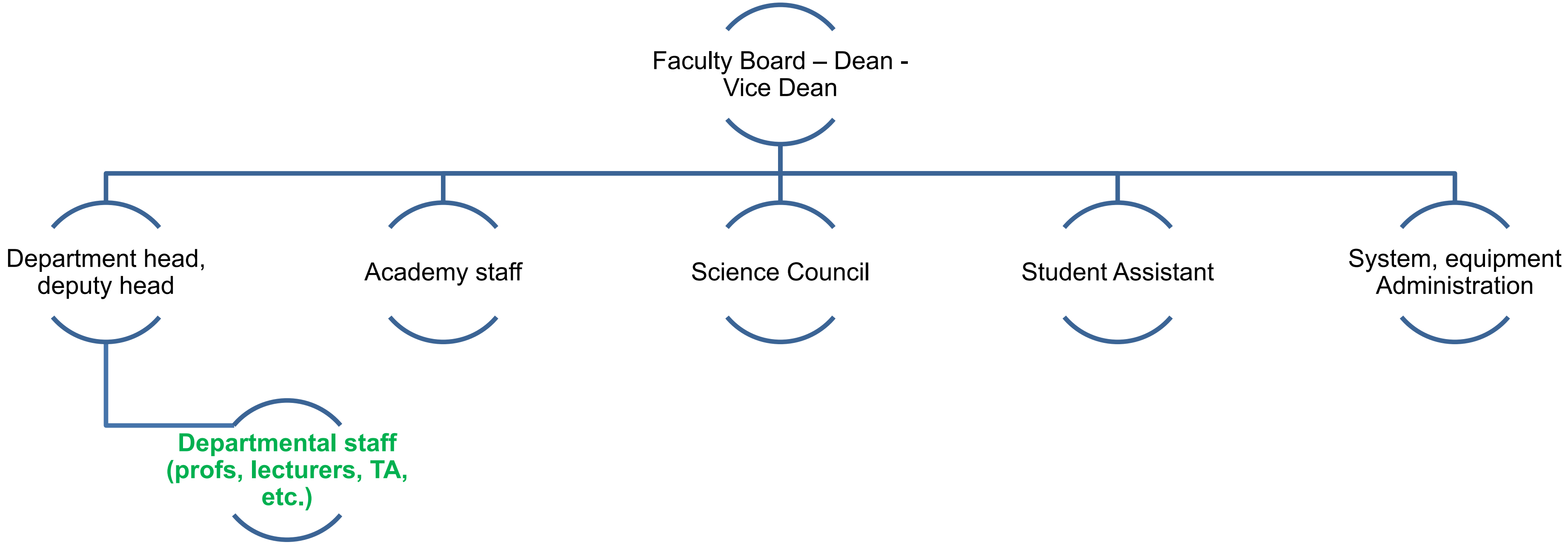
# Introduction to IT faculty

- **History**
  - Being one of the top 7 IT faculty of Vietnam
  - Established in 1995 with the precursor of the computer science department of the Faculty of Mathematics.



# Introduction to the IT faculty

- Organizational chart



# Department of Information Systems

- Providing the knowledge for students to implement IS projects, specialized in business applications and data management, etc.
- Focusing on applied technologies in the field of IS such as data modeling, database design, methods of analyzing and designing IS, distributed information systems.
- Research fields:
  - Information retrieval: Text Mining, Indexing Models, Natural Language Processing for IR, Cross Language IR, Question & Answers Systems, Medical Information Retrieval - (Leaded by prof. Ho Bao Quoc and Dr. Le Nguyen Hoai Nam)
  - Mobile IS: Cooperative Caching Method, Location Management, Mobile Databases, Mobile P2P Applications. Internet of Things –(Leaded by Dr. Nguyen Tran Minh Thu)
  - TEL systems, knowledge representation, semantic web, smart services (in education and business) – (Leaded by Dr. Pham Nguyen Cuong)
  - Information and data security: Data Confidentiality, Access Control Mechanisms, User Privacy, Data Privacy – (Leaded by Dr. Pham Thi Bach Huệ)
  - Data mining: Data mining in biomedicine – (Leaded by Dr. Lê Thi Nhàn)

# Department of Software Engineering

- Providing knowledge in the development, management and maintenance of software, thereby enabling students to design and implement high quality software products. Graduates of this major will be able to analyze, design, and administrate middle to high-end software projects
- Research fields:
  - Advanced methods in software design, object-oriented programming, teaching support software, cryptography and applications

# Department of Computer Networks and Telecommunications

- Providing knowledge in the field of communication between wide area networks, local computer networks and between distributed information systems. Student has a ability to design and implement medium to large computer networks, and communication systems
- Research fields:
  - Advanced network and communication technologies, distributed systems, VoIP, WAP / PKI systems and network security, mobile agent

# Department of Computer Science

- Providing advanced knowledge and skills necessary to build integrated applications that are capable of intelligent processing applied in education, training, economics, society, science and technology, natural resource and environmental management. Knowledge includes the foundation of knowledge based systems, human-to-machine interaction, pattern recognition, and data mining.
- Research fields
  - Knowledge base systems, soft calculation, image processing, neural networks, machine learning, pattern recognition, programming evolution, image and signal processing in medicine, semantic web, audio processing

# Department of Computer Vision

- Digital image and video processing, computer vision and robotics

# Programs

- Doctor of Philosophy in Computer Science (PhD)
- Master of Science in Computer Science or in Information Systems
- Bachelor of Science in Computer Science
  - Regular fulltime program (1995 – present)
  - Talented/honors fulltime program (2002 - present)
  - Advanced fulltime program (2013 - present )
  - English advanced fulltime program (2006 - present)
  - French advanced fulltime program (1994 - present); from 2010 to double degree with Claude Bernard University - Lyon1 (France)
  - Distance learning program (2006 - present )

# Campuses

- Campus 1 – 227 Nguyễn Văn Cừ, Chợ Quán Ward, HCMC
  - Specialized and Postgraduate studies;
  - >50 classrooms, 2 large lecture halls.
  - Equipped with sound systems & projectors;
  - Campus-wide WiFi coverage.
- Campus 2 – VNU-HCM Area, Đông Hòa Ward, HCMC
  - General studies;
  - >60 classrooms; 1 lecture hall with >400 seats.
  - Spacious campus, integrated infrastructure, sports and entertainment facilities.
  - https://www.youtube.com/watch?v=udb1I42NeNk

# Campus 1


SAB
SƠ ĐỒ TRƯỜNG (CƠ SỞ 1)
GĐ 2
F
H
C
B
A
E
D
I
GĐ 1
Phòng Bảo vệ
Trung tâm tin học
CỔNG CHÍNH
CỔNG VÀO BÃI GIỮ XE
Đường Nguyễn Văn Cừ
ĐẠI HỌC SƯ PHẠM TP.HCM
1 Hội trường I, tầng trệt tòa I
2 I.53, Văn phòng Khoa CNTT, tầng 5 tòa I
3 A.02, Phòng Công tác Sinh viên
4 B.01, Phòng Kế hoạch - Tài chính
5 B.02, Phòng Đào tạo
6 F.106, Văn phòng Đoàn
7 Thư viện, tầng 9-10 tòa I
WC Nam
WC Nữ
Khu tự học
KV lấy nước uống
Canteen
Agribank Autobank
E.001 , Trạm y tế
Quầy giáo trình
SƠ ĐỒ CƠ SỞ PHƯỜNG CHỢ QUÁN
Thực hiện bởi: Ban hoạt động sinh viên chương trình đề án (SAB)

# Computer Labs

- **09 computer labs** with internet connection, equipped with sound systems, projectors, and screens, with over 425 high-performance computers.
- Open on a rotating basis for self-study; pre-installed with teaching software.
- The computers are regularly maintained, upgraded annually, and replaced after 5 years of use.

# GPU Servers

- **04 high-performance GPU server systems**
  - Apollo ProLiant XL270d Gen10,
  - NVIDIA DGX A100,
  - DGX Station V100.

- With powerful configurations (multi-core CPUs, 256–512GB RAM, large-capacity A100/V100 GPUs), they optimally meet the needs for teaching and research in artificial intelligence, machine learning, and big data processing.

# Laboratories

- The Faculty has invested **over 70 billion VND** to build **02 laboratories**: the Artificial Intelligence Application Research Laboratory (AILab) and the Intelligent Systems Laboratory, equipped with a powerful GPU infrastructure featuring 32 NVIDIA A100 and V100 cores.
- These resources are open to faculty and students for research and can be registered for use. This provides a solid foundation for students and faculty of the IT Faculty to access advanced technology, enhancing the quality of training and scientific research.

# Library

- With over 19,000 book titles, more than 1,800 theses and dissertations, and a system of over 110 computers for research and study, along with nearly 500 seats, the library is a modern academic space that meets the research and study needs of faculty and students.

# Faculty Library

- Provide specialized resources in the field of Information Technology
- Support the learning, teaching, and research needs of students and lecturers of FIT
- Library book catalog: https://link.hcmus.edu.vn/Danhmucsach
- Book borrowing and returning regulations: https://link.hcmus.edu.vn/Quydinh
- Book borrowing registration: https://link.hcmus.edu.vn/Muonsach

# Systems

- Moodle
  - Learning materials
  - Assignments
  - Course-related announcements
  - Course discussion forum
  - Academic’s announcements
  - https://courses.fit.hcmus.edu.vn/

# Systems

- Q&A FIT
  - Students submit questions and inquiries
  - Support Teams provide responses
  - Ensuring the accuracy of information delivered
  - https://courses.fit.hcmus.edu.vn/q2a/